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Investigation on the influence of CH₄ hydrate dissociation on depressurization replacement efficiency in CO₂ hydrate sequestration

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Abstract

The replacement of CH₄ with CO₂ in marine hydrates has garnered widespread attention due to its potential for energy recovery and CO₂ sequestration. In this study, CH₄ hydrates and replacement process and outcomes with CO₂ were investigated using optical microscopy, differential scanning calorimeter (DSC), and Raman spectroscopy. Optical microscopy experiments showed that CH₄ hydrate formed a solid spherical hydrate film with a thickness of 5–10 μm on the surface of silica gel (SG), which arose from volume expansion associated with the water-to-hydrate phase transition. DSC experiments showed that CO₂ hydrate formation occurred during pure CH₄ decomposition, thereby reducing required CO₂ partial pressure. Finally, replacement ceased when the partial pressure of CO₂ below the partial pressure of CO₂ of the equilibrium pressure of the CH₄-CO₂ mixed gas hydrate. Pure CH₄ hydrate decomposition simultaneously achieved higher CH₄ recovery (87.9 mol%), CH₄ purity (84.7 mol%), and final system pressure (3.8 MPa). Due to CH₄ hydrate decomposed during replacement procedure (DSC results), the particle size (25–635 μm) minimally affected the replacement rate and efficiency, indicating that the replacement reaction was governed by reaction kinetics in SG. Raman analysis revealed consistent replacement efficiency across hydrate particle depths (0–60 μm).

Keywords CH₄-CO₂ replacement, Hydrate, Silica gel, Replaced reaction kinetic, Decomposition

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1 Introduction

Natural gas hydrates (NGHs) are non-stoichiometric crystalline compounds consisting of different hydrogen-bonded cages formed from water, in which gas molecules are encapsulated [1]. Gas species that can be encapsulated in cages with different structures under low-temperature and high-pressure conditions include methane (CH_4), carbon dioxide (CO_2) and others [2–4]. NGHs are considered prospective mineral energy reserves because the total CH_4 storage in CH_4 hydrate is approximately $3 \times 10^{15} \text{ m}^3$ at standard temperature and pressure (STP), and the amount of stored energy is estimated to be twice that of conventional fossils. Consequently, extensive investigations on exploiting these abundant resources have been conducted using different production methods [5–7]. To date, these methods are mostly based on thermodynamic principles, including depressurization [8–10], heat stimulation [11–13], and inhibitor injection [14]. However, these traditional methods result in NGHs decomposition for CH_4 production, which changes the mechanical and hydraulic properties of hydrate-bearing sediments (HBSs) [15, 16]. Consequently, during the exploitation of CH_4 hydrate using the depressurization or thermal stimulation methods, potential hazards such as submarine landslides and CH_4 leakage may occur [17].

Because of significant emissions of CO_2 , it has been identified as a key greenhouse gas that contributes to global warming [18]. Furthermore, CH_4 has a greenhouse effect approximately 26 times that of CO_2 . If CH_4 is accidentally released into the atmosphere during the extraction and decomposition of NGHs, it will exacerbate the greenhouse effect [19]. The CO_2 replacement method for extracting CH_4 hydrate not only mitigates the risks of submarine landslides and CH_4 leakage but also offers the added advantage of CO_2 sequestration [20, 21]. The heat released during the formation of CO_2 hydrates can offset the heat absorbed during the decomposition of seafloor NGHs [22]. Consequently, Research has been conducted on the sequestration of CO_2 and replacement of CH_4 within the hydrate layers [23]. These studies indicate that the process of replacement could prevent geological hazards [24]. Besides, the releasing heat of CO_2 hydrate formation (57.98 kJ/mol) is above absorbing heat of CH_4 hydrate decomposition (54.49 kJ/mol) if the releasing heat was fully absorbed by NGHs [25]. Therefore, the replacement method is promising for NGHs exploitation.

The CO_2 replacing NGHs method has been widely documented to explain its kinetics and mechanisms, including its physicochemical and thermodynamic controls [26–28]. The tight shell of the CO_2 - CH_4 mixed hydrate significantly impeded the diffusion of CO_2 and CH_4 [29, 30]. To enhance the diffusion of CO_2 , liquid CO_2 [31–33], CO_2/N_2 mixed gas [34, 35], and supercritical CO_2 [36]

had been tested in replacement experiments. However, the CO_2 sequestration fraction reported in the literature is less than 5%, whereas the proportion of extracted CH_4 gas is less than 20%, which severely limits the exploitation of NGHs using the replacement method [37]. Due to the challenge in transferring CO_2 or CH_4 from the gas phase to the water surface, the termination process of replacement may be controlled by diffusion within the quartz sand [38, 39]. Therefore, studying the reaction kinetics of replacement completion in quartz sand is extremely difficult [40, 41].

Notably, the decomposition of hydrates significantly affects the diffusion of CO_2 and CH_4 [42]. Computational and Raman spectroscopy studies have provided evidence that the initial rapid reaction stage may not involve inter-cage exchange but rather full decomposition of CH_4 hydrate and subsequent reformation of CO_2 hydrate [43–45]. However, the other instances of spontaneous replacement without hydrate decomposition have been documented in CO_2 - CH_4 systems based on experimental evidence [40, 46–49]. The deviation in decomposition latent heat between CO_2 hydrate and CH_4 hydrate is only 6% [25]. When the replacement rate is low or the time difference between CH_4 hydrate decomposition and CO_2 hydrate formation is small, the heat flow changes during the replacement process may not be detectable [50, 51]. Due to the difference in compressibility factors (Z) between CH_4 and CO_2 [52, 53], pressure changes occur during the replacement process.

Therefore, by simultaneously analyzing the heat flow and pressure variations, the time phases of the replacement process can be determined. However, no study has simultaneously analyzed the changes in pressure and heat flow during the replacement process to determine if hydrate nuclei break during the replacement.

In-situ combustion can effectively provide CO_2 for CH_4 hydrate layers. However, due to the hazards associated with high-pressure combustion, there are currently no experimental studies on the efficiency of in-situ combustion for CH_4 hydrate extraction. Based on the obstacles to CH_4 hydrate extraction (diffusion resistance, heat absorption during decomposition, pressure/production rate decline), it is advisable to first conduct simulated in-situ combustion experiments to analyze the most efficient combustion methods.

A literature review has revealed that no study has explored pure CH_4 hydrate decomposition's effect on CH_4 - CO_2 replacement factor ($R.F.$) in micron-sized pure spherical hydrate particles. Similar to the separation coefficient used in the hydrate method for separating biogas, we can simultaneously evaluate the separation effects of both CH_4 and CO_2 [54, 55]. Likewise, the replacement coefficient can concurrently measure the sequestration

efficiency of CO₂ and the extraction efficiency of CH₄. While Ouyang et al. [56] enhanced CH₄ proportion in the residual gas ($y_g^{CH_4}$) to 79.1 mol% and sequestration fraction of CO₂ (S_{Fr,CO_2}) to 88.2% through multi-step depressurization, the final pressure remained low (1.99 MPa), leaving the efficiency mechanism unclear. Recent research highlights gas partial pressure's role: Gambelli [57] observed CH₄ recovery dropping from 84.14% to 21.54% as CO₂ concentration increased from 67.55% to 91.83%, suggesting excessive CO₂ partial pressure ($P_g \cdot CO_2$) inhibits CH₄ release and requires optimization. Li et al. [58] found excessive CH₄ partial pressure ($P_g \cdot CH_4$) occupies hydrate surface sites, hindering CO₂ adsorption, requiring optimization via partial pressure ratio ($P_g \cdot CO_2 / P_g \cdot CH_4$), and replacement efficiency increased significantly when this ratio exceeded 1.2.

In the quartz sand system, due to the resistance to gas diffusion, there have been no research reports on the highest $P_g \cdot CH_4$ and lowest $P_g \cdot CO_2$ achievable during the replacement process. This gap has also resulted in no research reporting the impact of CH₄ hydrate decomposition on $P_g \cdot CO_2$ and the influence of CO₂ hydrate formation on $P_g \cdot CH_4$.

Spherical silica gel (SG) is an ideal porous structure for the quick formation of spherical CH₄ hydrate particles, as there is no difficulty in transferring CO₂ from the gas phase to the hydrate surface [59, 60]. Therefore, spherical CH₄ hydrate particles are conducive to studying the influence of CH₄ hydrate decomposition on the ultimate partial pressures of CH₄ and CO₂. Meanwhile, SG embedded with spherical CH₄ hydrate particles is a useful model for studying the most efficiency of the replaced reaction of combustion methods. Increasing the diameter of CH₄ hydrate particles helps to enhance the time difference between CH₄ hydrate decomposition and CO₂ hydrate formation, which facilitates the detection of thermal flow changes during the replacement process.

To address the aforementioned gaps, in this study, silica gels (SGs) with different particle sizes were used to CH₄ hydrates at 100% water saturation in the pore, resulting in a 100% conversion rate of water [61, 62], to analyze the effect of hydrate particle size on replacement efficiency. The effect of pure CH₄ hydrate decomposition and particle size on the CO₂ sequestration fraction,

the CH₄ recovery fraction, and the efficiency of replacement was investigated. The effect of $P_g \cdot CH_4$ and $P_g \cdot CO_2$ on replacement reaction was analyzed. The CH₄ fraction in the residual gas was confirmed by gas chromatography (GC). Raman spectroscopy was employed to measure the cage occupancy in hydrates and the replacement efficiency at different positions and within the inner part of the spherical CH₄ hydrate particles. An in situ stereoscopic microscope was used to analyze the particle surface, shape, and thickness of the hydrate layer. A stereoscopic microscopy was used to analyze the hydrate in the pore volume of the SG. The heat flow during the replacement process was investigated using a high-pressure micro-differential scanning calorimeter (HP DSC) with a quantitative CO₂ injection. By combining pressure and heat flow information to determine the time period of the replacement reaction, and further investigating whether CH₄ hydrate undergoes decomposition during the replacement process.

2 Experimental methodology

2.1 Materials

High-purity CH₄ and CO₂ (99.99%) from Huatpeng Special Gases Co., Ltd., China, were used in this study. Spherical SG particles with a pore diameter of 77.14 nm and size ranges of 25–143 μm (SG I), 47–302 μm (SG II), and 112–635 μm (SG III), sourced from Silicycle, Canada, were utilized to simulate the pore structure of sandy sediments in the South China Sea. The textural properties of SGs are in Table 1. Deionized water (18 mΩ·cm resistivity) prepared in-house was used for all experiments.

2.2 Apparatus

Figure 1 illustrates the hydrate replacement apparatus featuring dual 316L stainless steel reactor (R1: 149.5 mL) rated for 20.0 MPa with internal roughness < 0.2 μm. Each reactor resided in independently temperature-controlled baths (233.1–333.1 K ± 0.1 K), equipped with PT100 thermometers (± 0.1 K) and CYB-20S pressure sensors (± 0.02 MPa). Gas mass flow was monitored by Bronkhorst H-M13 meters (0–1.5 kg/h, ± 0.5%) at inlet/outlet ports. An Agilent data collector recorded parameters at 10-s intervals through integrated transducers.

Table 1 Textural properties of the SG

No	Silica gel 1 (SG I)	Silica gel 2 (SG II)	Silica gel 3 (SG III)
Mean particle diameter (μm)	25–143	47–302	112–635
Mean pore diameter (nm)	77.14	77.14	77.14
Specific pore volume (m ³ /kg)	7.5 × 10 ⁻⁴	9 × 10 ⁻⁴	7.9 × 10 ⁻⁴
Specific surface area (m ² /kg)	42.8 × 10 ³	48.0 × 10 ³	42.7 × 10 ³

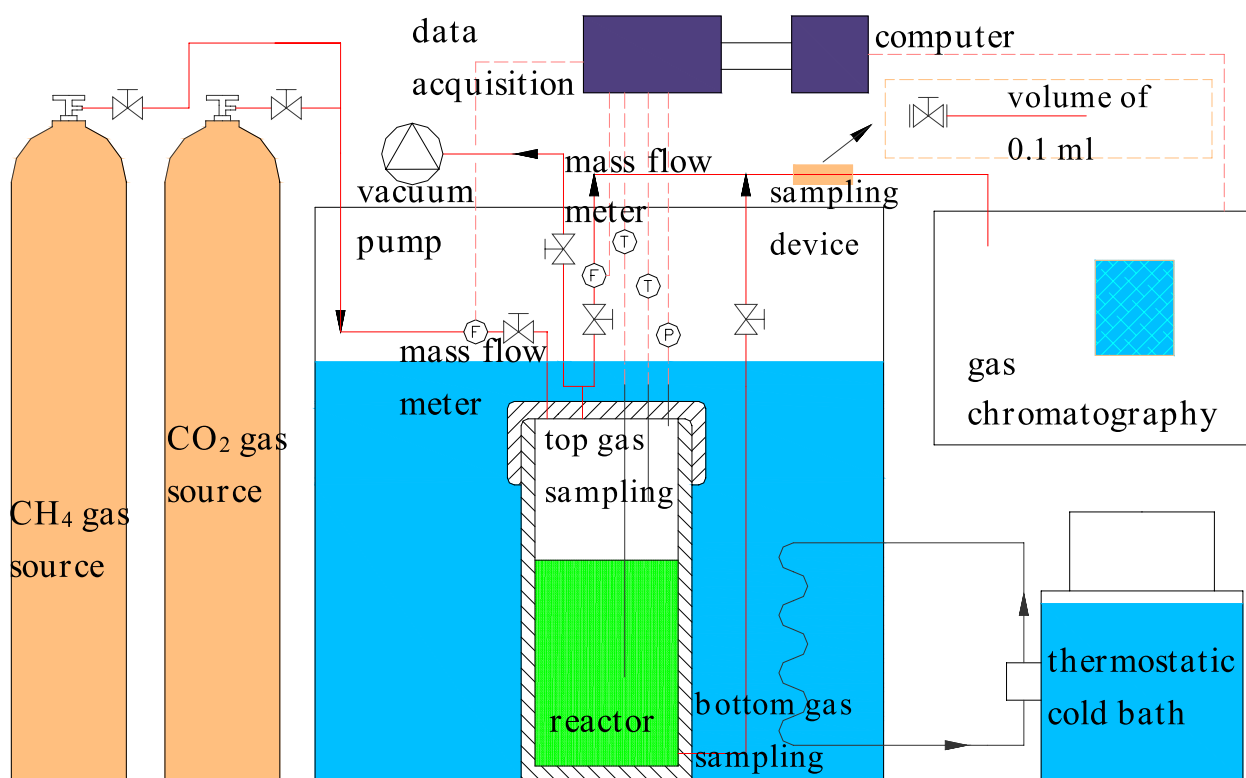


Fig. 1 Schematic diagram of the CH₄ hydrate formation and CO₂-CH₄ replacement apparatus used in this work

To enable real-time gas analysis during hydrate replacement, install a ball valve before and after a 0.1 ml steel tube sampler. Connect a steel tube sampler to both the top and bottom of the reactor. Open the reactor valves and the valves connected to the samplers. Once the steel tube samplers are filled with high-pressure sample gas, close the reactor and sampler valves, remove the samplers from the reactor, connect them to the Trace 1300 gas chromatograph, and then analyze the sample proportions.

2.3 Experimental procedures

2.3.1 Characterization of SG

SG I, II and III diameters were analyzed using a laser particle analyzer for CH₄ hydrate particles. Their pore size and volume were measured via mercury intrusion porosimetry. The mercury porosimeter model used is the MicroActive AutoPore V 9600, version 2.03.00. Both the advanced contact angle and the recommended contact angle employed for the test were set at 130.000°. The temperature of the mercury was maintained at 301.3 K throughout the testing process. SG morphology was characterized by a field emission scanning electron microscopy (Cryo-FE-SEM, Hitachi S-4800), shown in previous studies [63, 64].

2.3.2 Preparation of SG bed

All SG types were vacuum-dried at 373.1 K overnight to remove free water. Water equivalent to the SG pore volumes was added for saturation. SG was centrifuged at 3000 rpm for 3 min to disperse water uniformly, as shown previously [64].

2.3.3 CH₄ hydrate formation and CO₂-CH₄ replacement procedure

Table 2 shows CH₄ hydrate formation cases: 6 for R1. Reactors filled with saturated SG were sealed, pressurized to 1 MPa then depressurized three cycles. After vacuum evacuation to remove air, water withdrawal absence was confirmed through CH₄ consumption. Reactors were cooled to target temperatures (Table 2), then quantitatively measured CH₄ gas was injected using mass flow meter. CH₄ hydrate formation took more than 24 h. At the end, the pressure exceeded 10 MPa, and the pressure-temperature condition was maintained for more than 60 min.

Table 3 summarizes CH₄ hydrate replacement with CO₂ in 6 cases (R1). CH₄ hydrate was cooled to a temperature range of 233–243 K to prevent hydrate dissociation at atmosphere pressure, followed by CH₄ gas release monitored via outlet flow meters to the atmosphere. CO₂ was subsequently introduced into

Table 2 Summary of CH₄ hydrate formation experiments, including gas consumption, hydration number, water conversion, and so on

EXP. No	Beginning of CH ₄ hydrate formation experiment				End of CH ₄ hydrate formation experiment					
	The volume ratio of gas to water (L L ⁻¹)	SG	T (K)	Water saturation of pore (%)	molar amount of water in pore (mol)	P (MPa)	The consumed CH ₄ gas (mol)	Hydra- tion number	Fully Water conversion time (days)	Water conversion (mol%)
Case 1-R1-II	5.12	II	273.1	100	1.2556	15.7	0.2152	5.83	1	100
Case 2-R1-II	5.12	II	273.1	100	1.2556	12.3	0.2025	6.20	1	100
Case 3-R1-II	5.12	II	273.1	100	1.2556	12.7	0.2028	6.19	1	100
Case 4-R1-II	5.12	II	273.1	100	1.2556	15.6	0.2113	5.94	3	100
Case 5-R1-III	5.04	III	273.1	100	1.2556	16.7	0.2048	6.13	9	100
Case 6-R1-I	4.97	I	273.1	100	1.2556	16.7	0.2125	5.91	0.4	100

Table 3 Summary of CH₄ hydrate replaced by CO₂ experiments, including replacement time, CH₄ proportion in residual gas, and so on

EXP. No	Beginning of replacement experiment			End of replacement experiment					
	Injection gas (mol)	N ₂ proportion (mol%)	The mole ratio of injection gas to water (mol mol ⁻¹)	P (MPa)	T (K)	Replace-ment time (h)	T ₉₀ ^a (h)	Top CH ₄ proportion in residual gas (mol%)	Bottom CH ₄ proportion in residual gas (mol%)
Case 1-R1-II	0.0954	0	0.0760	3.6	278.1	120	7.3	73.54	73.77
Case 2-R1-II	0.0752	0	0.0599	3.7	278.1	216	6.0	80.52	80.71
Case 3-R1-II	0.0523	0	0.0416	3.8	278.1	120	3.5	84.67	84.86
Case 4-R1-II	0.2288	0	0.1823	4.2	278.1	72	6.3	52.03	52.74
Case 5-R1-III	0.2291	0	0.1825	4.0	278.1	168	16.4	53.73	53.68
Case 6-R1-I	0.2284	0	0.1819	4.1	278.1	192	13.4	55.34	55.62

^a T₉₀ was defined as the time required for the replacement of CH₄ gas to reach 90% of the total gas replaced

reactors using inlet flow meters, with monitored temperature and pressure measurements which confirmed the absence of thermal dissociation throughout the injection process. The replacement process lasted 7–9 days. During this period, 10–12 high-pressure gas samples of 0.1 mL were collected simultaneously from the top and bottom of the reactor, and the CO₂/CH₄ proportions in the samples was analyzed using GC. When this ratio stabilized, indicating replacement completion.

Thirdly, CO₂-CH₄ hydrate was cooled to a temperature range of 233–243 K to prevent hydrate dissociation and CO₂ liquefaction at atmosphere pressure. The pressure of the reactor was then reduced to atmospheric pressure. The residual CO₂-CH₄ mixed gas in R1 was collected using sampling bag, and monitored via outlet mass flow meter. GC then confirmed gas proportions matching prior sampling ratios.

After releasing residual CO₂-CH₄ gas, R1 was heated to 300 K to dissociate all mixed gas hydrate. Table 4 details CO₂-CH₄ hydrate decomposition in 6 cases (R1). Dissociated gas volume was measured via mass flow meter, and gas ratio analyzed with GC.

2.3.4 Hydrate characterization by Raman, optical microscopy, laser particle analyzer and HP-DSC for saturated SG

An optical microscopy was used for in-situ analysis of CH₄ hydrate formation locations in SG III and the thickness of hydrate layers on the surface of SG III in a sapphire-window reactor. After multiple rounds of analysis and calculation, the average thickness of the CH₄ hydrate layer formed on the surface of 16 SG II particles was determined to obtain a mean value for comparison with theoretical values. A 24× stereo microscope was used to analyze CH₄ hydrate decomposition on the surface of SG II at 293.1 K and 15 MPa.

In Raman experiments, water-saturated SG III and SG II formed CH₄ hydrate in reactor R1. CO₂ replacement was conducted under conditions matching cases 4 and 5. The CH₄/CO₂ mixed gas hydrate and CH₄ hydrate were transferred to a liquid nitrogen-cooled plate for Raman analysis at 213.1 K. Spectra (1 cm⁻¹ resolution) were obtained using a HORIBA XploRA system with 1800 grooves/mm grating and 532 nm/50 mW laser.

A high-pressure micro-differential scanning calorimeter apparatus (HP DSC BT2.15 Evo, Setaram Inc., France) was employed to measure CH₄ hydrate formation and

Table 4 Summary of the decomposition of mixed CO₂-CH₄ hydrate, moles of decomposition gas, CH₄ proportion

EXP. No	Decomposition of mixed CO ₂ -CH ₄ hydrate						Residual gas
	P (MPa)	T (K)	Moles of decomposition gas (mol)	Hydration number	CH ₄ proportion (mol%)	CO ₂ proportion (mol%)	Mean CO ₂ proportion (mol%)
Case 1-R1-II	2.2	305.1	0.1153	-	56.02	43.98	25.75
Case 2-R1-II	1.5	298.1	0.0840	-	58.75	41.25	14.92
Case 3-R1-II	1.0	298.1	0.0625	-	65.48	34.52	13.32
Case 4-R1-II	3.8	303.1	0.2052	6.12	41.23	58.77	47.61
Case 5-R1-III	3.6	300.1	0.1853	6.77	43.44	56.56	48.56
Case 6-R1-I	3.5	298.1	0.1940	6.47	39.36	60.64	43.23

decomposition. Temperature was controlled via liquid nitrogen. During CO₂-CH₄ replacement in SGII hydrate, both heat flow and pressure data were recorded. The system contains two Hastelloy C276 cells (3.6 mL each, 40.0 MPa pressure capacity), operating from 213.15 K to 393.15 K with 0.02 μW resolution and ± 0.2 K accuracy.

Table S1 details five DSC experiments. In each, 0.99 g of water-saturated SGII (25.98 mmol H₂O) filled 30% of 3.6 mL cell (R2). The reference cell stayed empty. After air purging, both cells cooled to 253.1 K (atmospheric pressure) to freeze water. At 293.1 K, CH₄ gas was employed to pressurize R2 to 15 MPa. Subsequently, the system was cooled to 253.1 K (0.1 K/min) to facilitate the formation of hydrates and ice. At a pressure of 12 MPa, R2 was cooled from 293.1 K to 273.1 K (0.1 K/min) and then maintained for 24 h to ensure complete conversion of water to hydrate. Subsequent cooling to 253.1 K confirmed no ice formation. Finally, heating to 298.1 K (0.1 K/min) decomposed the hydrate.

Following complete water-to-CH₄ hydrate conversion in R2, the system was cooled to 223.1–233.1 K to release residual CH₄ to atmospheric pressure. subsequently, CO₂ was then injected quantitatively using a mass flow meter. The temperature was increased to 278.1 K (0.1 K/min) and maintained for 24 h for CH₄-CO₂ replacement. Finally, heating to 298.1 K (0.1 K/min) for 1 h decomposed the CO₂-CH₄ mixed hydrate.

2.4 Numerical calculations

2.4.1 CH₄ consumed (Δn^{CH₄})

The CH₄ gas amount pressurized into the reactor, denoted $n_0^{CH_4}$, was calculated using Eq. (1):

$$n_0^{CH_4} = \frac{m_0^{CH_4}}{M^{CH_4}} \quad (1)$$

Here, $m_0^{CH_4}$ corresponds to the quantified CH₄ mass determined via mass flow meter measurements throughout the CH₄ injection procedure, where M^{CH_4} represents the molar mass of CH₄.

After CH₄ hydrate formation, the CH₄ consumed was calculated by subtracting the residual gas moles measured by a gas meter from the initial moles pressurized into the reactor, using Eq. (2) to determine Δn^{CH₄}.

$$\Delta n^{CH_4} = n_0^{CH_4} - \frac{m_E^{CH_4}}{M^{CH_4}} \quad (2)$$

Here, $m_E^{CH_4}$ was CH₄ mass measured via the mass flow meter during release, and $n_0^{CH_4}$ was calculated using Eq. (1).

The hydration number in Table 2 reflects the molar ratio of consumed gas to water during formation, with calculation details provided in the previous study [64].

2.4.2 Split fraction and replacement factor

The quantities of CO₂ initially introduced under pressure into the reactor vessel, designated as $n_0^{CO_2}$, was determined through application of Eqs. (3) and (4):

$$n_0^{CO_2} = \frac{m_0^{CO_2}}{M^{CO_2}} \quad (3)$$

Here, $m_0^{CO_2}$ represents the mass of CO₂ measured through the mass flow meter during the CO₂ injection process. M^{CO_2} denotes the molar mass of CO₂.

At the conclusion of the CH₄-CO₂ replacement, the moles of CH₄ and CO₂ gases present in the reactor's residual gas were designated as $n_g^{CH_4}$ and $n_g^{CO_2}$, respectively, and these were calculated using Eqs. (4), and Eq. (5).

$$n_g^{CH_4} = \frac{m_g}{M_g} * y_g^{CH_4} \quad (4)$$

$$n_g^{CO_2} = \frac{m_g}{M_g} * y_g^{CO_2} \quad (5)$$

Here, m_g and M_g represent the mass and molar mass of the residual gas, respectively, as measured through the mass flow meter during the release process. Additionally, $y_g^{CH_4}$ and $y_g^{CO_2}$ denote the CH₄ and CO₂ proportions in the residual gas, respectively, which were determined by analyzing the gas sample from a sample bag.

Following the dissociation of the CH₄-CO₂ mixed hydrate, the moles of CH₄ and CO₂ gases present in the dissociated gas within the reactor were represented as $n_H^{CH_4}$ and $n_H^{CO_2}$, respectively, and their quantities were calculated using Eq. (6) and Eq. (7).

$$n_H^{CH_4} = \frac{m_H}{M_H} * y_H^{CH_4} \quad (6)$$

$$n_H^{CO_2} = \frac{m_H}{M_H} * y_H^{CO_2} \quad (7)$$

Here, m_H and M_H represent the mass and molar mass of the dissociation gas of hydrate, respectively, as measured through the mass flow meter during the release process. Additionally, $y_H^{CH_4}$ and $y_H^{CO_2}$ denote the CH₄ and CO₂ proportions in the dissociation gas of hydrate, respectively, which were determined by analyzing the gas sample from a sample bag.

$S.Fr.CO_2$ in the hydrate for quantifying the selectivity of the hydrate was the CO₂ captured in hydrate to CO₂ supplied, calculated using Eq. (8):

$$S.Fr.CO_2 = \frac{n_H^{CO_2}}{n_0^{CO_2}} \quad (8)$$

The recovery fraction of CH_4 ($R_{\bullet}Fr_{\bullet}\text{CH}_4$) in the equilibrium gas for quantifying the selectivity of the equilibrium gas was CH_4 released in equilibrium gas to CH_4 consumed in CH_4 hydrate formation process, calculated using Eq. (9):

$$R_{\bullet}Fr_{\bullet}\text{CH}_4 = \frac{n_g^{\text{CH}_4}}{\Delta n^{\text{CH}_4}} \quad (9)$$

$R.F.$ for the CH_4 - CO_2 replacement, which simultaneously evaluates the efficiency of CH_4 hydrate exploitation and CO_2 sequestration, is defined analogously to the separation factor (S.F.) [54, 55]. The $R.F.$ was defined as the ratio of the moles of CO_2 to CH_4 in the hydrate phase to that in the gas phase, calculated employing Eq. (10):

$$R.F. = \frac{(1 - y_H^{\text{CH}_4}) \times y_g^{\text{CH}_4}}{y_H^{\text{CH}_4} \times (1 - y_g^{\text{CH}_4})} \quad (10)$$

Here, $y_g^{\text{CH}_4}$ denotes CH_4 proportion in residual gas, while $y_H^{\text{CH}_4}$ represents that in hydrate dissociation gas, both determined through sample bag gas measurements.

2.4.3 CH_4 and CO_2 partial pressure

Due to the differing Z values of CH_4 and CO_2 [52, 53], $P_g \cdot \text{CH}_4$ and $P_g \cdot \text{CO}_2$ should be calculated from CH_4 and CO_2 moles in residual gas, respectively, using Eqs. (11) and (12):

$$P_g^{i+1} \cdot \text{CH}_4 = P_g^i \cdot \text{CH}_4 * \frac{n_g^{\text{CH}_4}}{n_g^{i\text{-CH}_4}} \quad (11)$$

$$P_g^{i+1} \cdot \text{CO}_2 = P_g^i \cdot \text{CO}_2 * \frac{n_g^{\text{CO}_2}}{n_g^{i\text{-CO}_2}} \quad (12)$$

Here, $P_g^0 \cdot \text{CH}_4$ and $P_g^0 \cdot \text{CO}_2$ represented products of P with $y_g^{\text{CO}_2}$ and $y_g^{\text{CH}_4}$, respectively. These values were used to calculate $n_g^{0\bullet\text{CH}_4}$ and $n_g^{0\bullet\text{CO}_2}$ using gas state equation, respectively, which were less than $n_g^{\text{CH}_4}$ and $n_g^{\text{CO}_2}$, respectively. $P_g^1 \cdot \text{CH}_4$ and $P_g^1 \cdot \text{CO}_2$ were calculated using Eqs. 11 and 12, respectively, and were higher than $P_g^0 \cdot \text{CH}_4$ and $P_g^0 \cdot \text{CO}_2$.

$P_g^1 \cdot \text{CH}_4$ and $P_g^1 \cdot \text{CO}_2$ were used to calculate $n_g^{1\bullet\text{CH}_4}$ and $n_g^{1\bullet\text{CO}_2}$ using gas state equation, respectively, which were higher than $n_g^{0\bullet\text{CH}_4}$ and $n_g^{0\bullet\text{CO}_2}$.

Residual gas moles $n_g^{i\bullet\text{CH}_4}$ and $n_g^{i\bullet\text{CO}_2}$ were derived from $P_g^i \cdot \text{CH}_4$ and $P_g^i \cdot \text{CO}_2$, respectively, using iterative calculations for $n_g^{i\bullet\text{CH}_4}$ and $n_g^{i\bullet\text{CO}_2}$, etc. Where superscript i denotes iterations. After 3 iterations, $P_g^{i+1} \cdot \text{CH}_4$ approached $P_g^i \cdot \text{CH}_4$, and $n_g^{i\bullet\text{CH}_4}$ converged to $n_g^{\text{CH}_4}$, and the deviation between $n_g^{i\bullet\text{CH}_4}$ and $n_g^{\text{CH}_4}$ was less than 1%, similarly for CO_2 parameters.

3 Results and discussion

3.1 The CH_4 hydrate particles formed a dense shell on the SG

3.1.1 The analysis of the water conversion to hydrate

Figure 2a shows that median particle sizes were 68.3 μm (SG I), 112.2 μm (SG II), and 267.0 μm (SG III). Figure 2b shows that the median pore diameters of the three types of SG were all approximately 80 nm, with the specific pore volumes of SG I, SG II, and SG III being 0.75, 0.85, and 0.78 ml/g, respectively.

Firstly, water-to- CH_4 hydrate conversion reached 100% in all SGs, consistent with previous studies [61, 62] and DSC results (Fig. S1). Figure S1a demonstrates that water underwent phase transition to ice at approximately 262.5 K under atmospheric pressure within the silica gel matrix, with subsequent melting observed at 271.1 K.

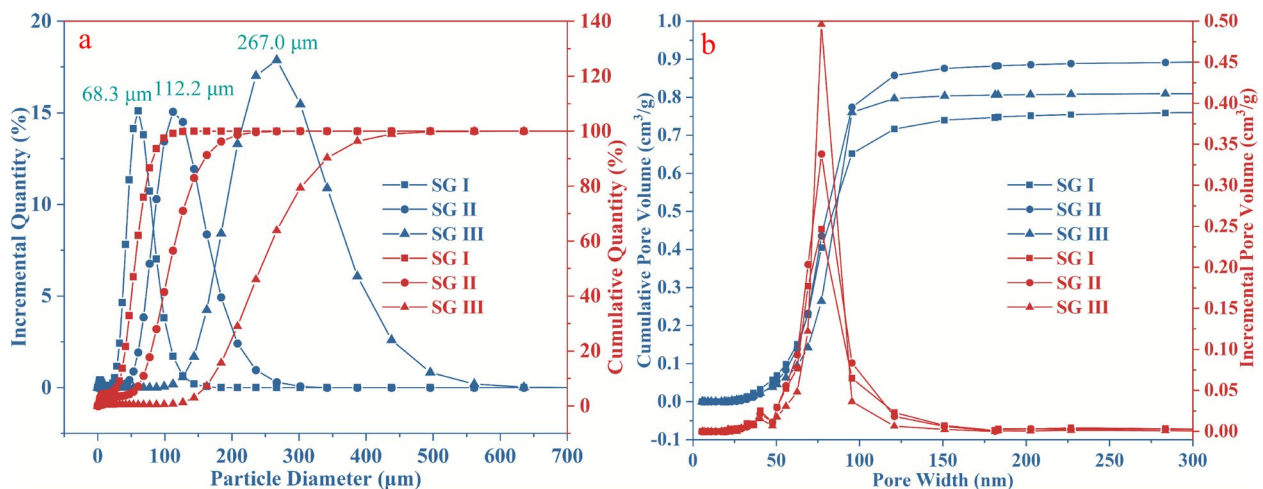


Fig. 2 The distribution diameter, pore volume and size of SGs. **a** distribution diameter; **(b)** pore volume and size

Figure S1b depicts the sequential phase behavior under controlled cooling conditions (0.1 K/min from 293.1 K to 253.1 K) at 12.0 MPa initial pressure, where partial water conversion to CH_4 hydrate occurred at 275.3 K followed by residual water crystallization at 261.8 K. Figure S1c demonstrates full hydrate formation achieved by maintaining isothermal conditions at 273.1 K for 24 h under 12.0 MPa final pressure. This conclusion is supported by the no detectable ice formation during subsequent cooling to 253.1 K, as indicated by thermal analysis. During the reheating phase to 298.1 K, only CH_4 hydrate dissociation endothermic peak was observed at 287.1 K. These results demonstrate that complete water-to-hydrate conversion necessitated maintaining pressures above 10.0 MPa combined with isothermal stabilization at 273.1 K for at least 24 h. Measured hydration numbers (5.75–6.20, Table 2) aligned with sI hydrate's stoichiometric 5.75 within experimental error [65].

3.1.2 The analysis of the thickness of the CH_4 hydrate layer covering the SG

Figure 3(a–p) shows the stereoscopic microscope images of porous SGs with 100% water converting to hydrate. Figures 3a–b reveal a 7.9 μm diameter increase in SG II after hydration, indicating a 3.9 μm thick hydrate layer. Experimental measurements of 16 CH_4 hydrate layers formed on SG II particles showed an average thickness of 3.05 μm (Fig. 4), which aligned closely with theoretical calculations (3.05 μm) derived from Equation (S1) using the established SG II particle diameter of 112.2 μm (Fig. 2a). Figures 3c–d show a 10.3 μm diameter expansion in SG III. Figures 3e–p illustrate water re-adsorption into SG pores during hydrate decomposition.

In a summary, volume expansion subsequently formed ice or hydrate shell layers outside SG spheres. The CH_4 hydrate particles formed a dense shell, hence the contact area between the hydrate phase and the gas phase remained constant during CO_2 injection, which is crucial for studying the intrinsic kinetics of replacement reactions.

3.2 Pure CH_4 hydrate decomposition facilitates CO_2 clathrate sequestration and replacement efficiency

3.2.1 Analysis of pure CH_4 hydrate decomposition amount

In cases 1–4 with water-saturated SGII filling one-third of R1, CH_4 hydrate decomposition's effect on $P_g \cdot \text{CO}_2$ was studied. After CO_2 injection, temperature/pressure peaked within 3–4 h (Fig. 5a/S2). In Case 2, the CH_4 proportions at the top and bottom rapidly increased to 65.9 mol% and 91.4 mol%, respectively, within 3–4 h, aligning with Cases 1/3 trends (Figs. 5b–d). Final CH_4 - CO_2 mixed hydrate decomposition measured 0.1153, 0.084, and 0.0625 mol respectively (Table 4),

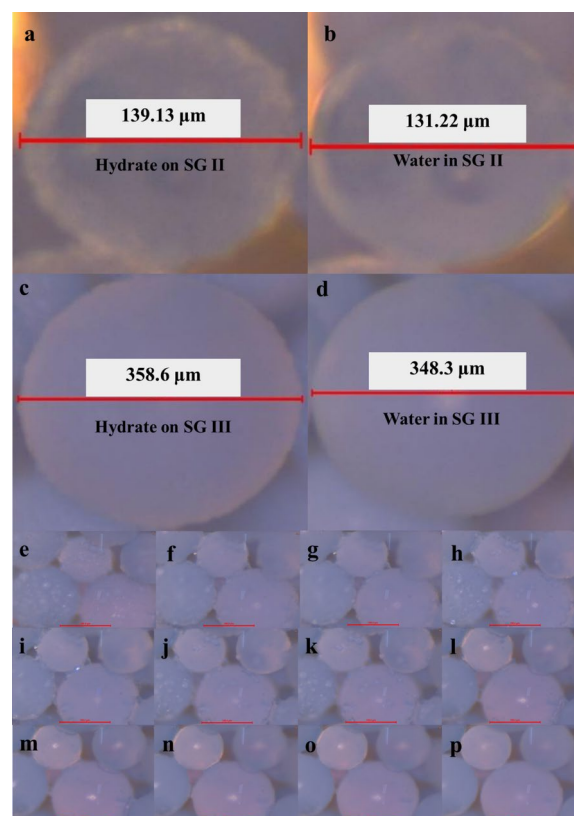


Fig. 3 Stereoscopic microscope images of porous SGs with 100% water converting to hydrate, are presented as follows: (a) Stereoscopic microscope image of SGII with 100% water converting to hydrate, (b) stereoscopic microscope image of SG II with 100% water, (c) stereoscopic microscope image of SG III with 100% water converting to hydrate, and (d) stereoscopic microscope image of SG III with 100% water, (e–p) CH_4 hydrate decomposition process

significantly lower than CH_4 hydrate formation levels (Table 2), indicating partial CH_4 hydrate dissociation occurred during the 3–4 h period.

Figure 5e indicates that lower CO_2 injection amounts led to higher pressure, residual gas CH_4 composition, and fewer moles of decomposed CH_4 - CO_2 hydrate in cases 1–3. The pure CH_4 hydrate decomposition yielded 0.0917, 0.1247, and 0.1518 mol CH_4 in cases 1, 2, and 3, respectively. Increased CO_2 injection replaced more CH_4 from hydrates, elevating residual gas $P_g \cdot \text{CH}_4$ and inhibiting further decomposition. Finally, the higher the $P_g \cdot \text{CH}_4$ was, the lower the $P_g \cdot \text{CO}_2$ in cases 1–3.

3.2.2 Analysis of the effect of pure CH_4 hydrate decomposition on replacement efficiency

The greater the amount of pure CH_4 hydrate decomposed, the higher the $P_g \cdot \text{CH}_4$, but both remained below the CH_4 hydrate P^{eq} in cases 1–4 (Table 5, Fig. 5f). The more the pure CH_4 hydrate decomposition, the higher

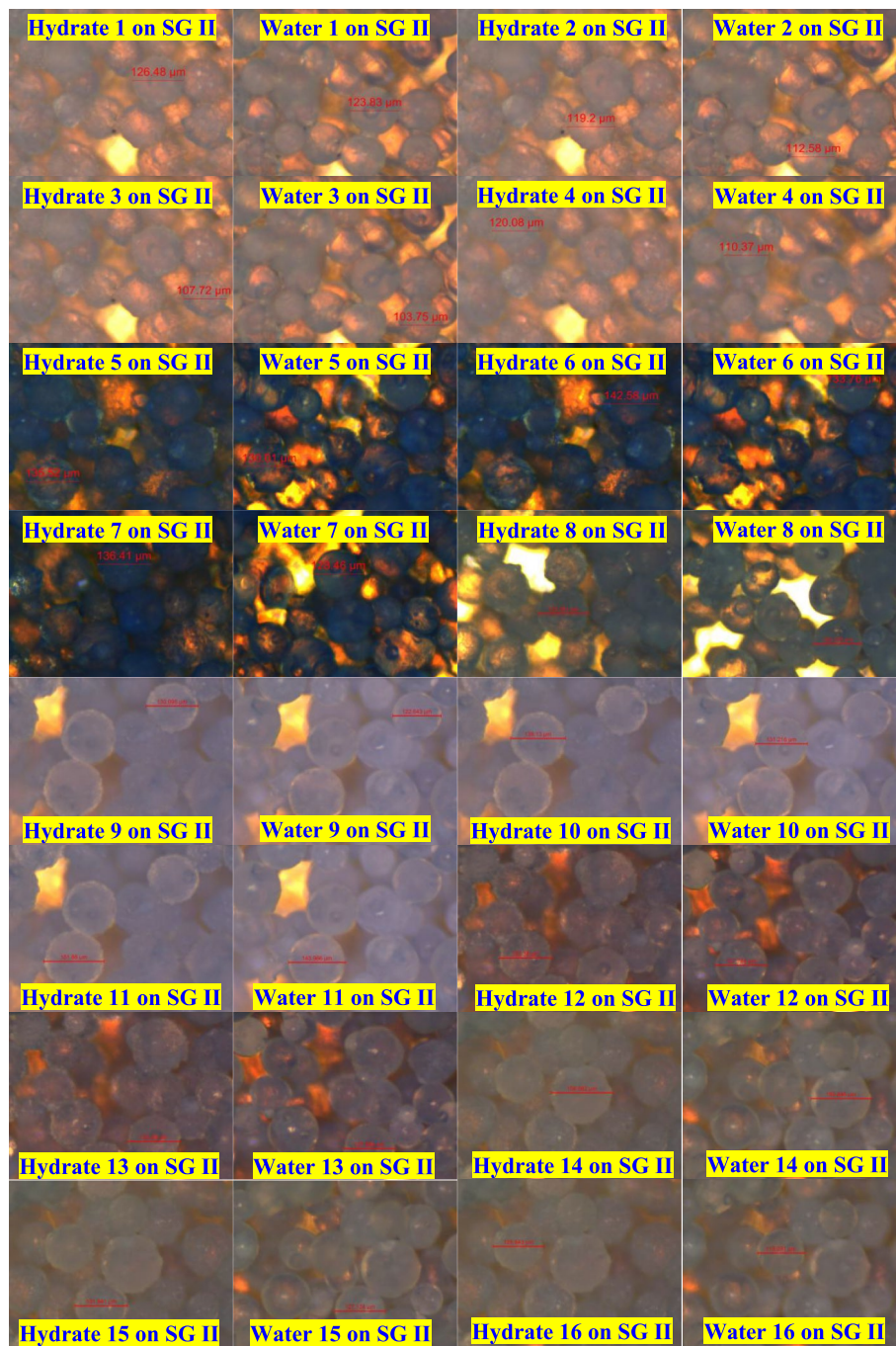


Fig. 4 Diagram showing changes in the diameter of water particles as they transform into hydrate particles

the $R.Fr.CH_4$ and CH_4 proportion in residual gas, the lower the $P_g \cdot CO_2$, and it had a minor impact on the $S.Fr.CO_2$ (Fig. 5f). $R.Fr.CH_4$ and CH_4 proportion in residual gas were improved to 87.9 mol% and 84.8 mol%, respectively. The $P_g \cdot CO_2$ decreased to 0.7 MPa.

Simultaneously, the decomposition of CH_4 hydrate caused an increase in the difference of CO_2 proportions

in the hydrate and the gas phase and the R.F., reducing the time required to complete 90% replacement (T_{90}) (Fig. 6a). The difference of CO_2 proportions in the hydrate and the gas phase and the R.F. were improved to 26.3 mol% and 6.03, respectively. The T_{90} decreased to 3.5 h.

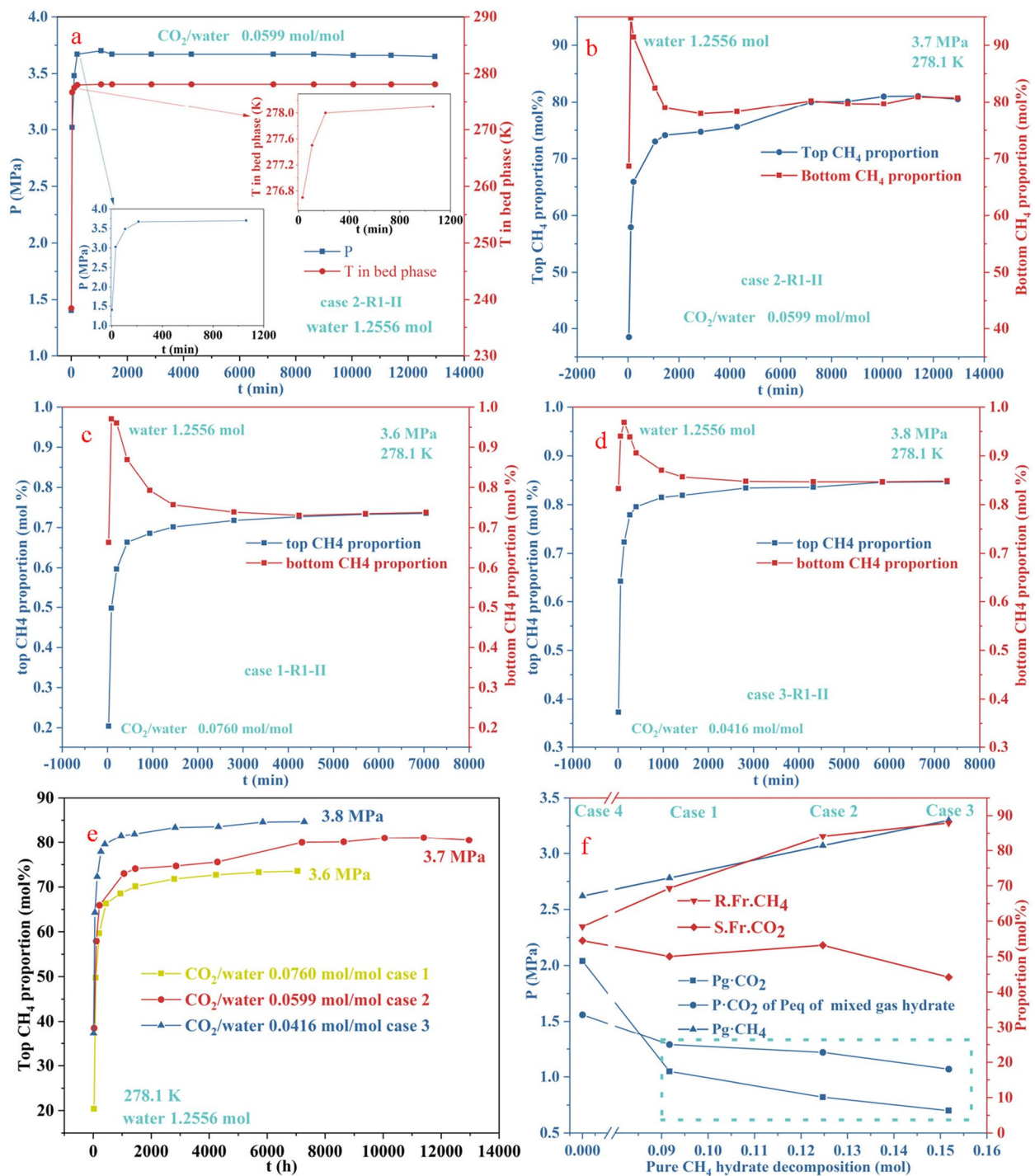


Fig. 5 The temporal variation of pressure, temperature, and the CH₄ compositions in the residual gas phase at the top and bottom at a constant temperature of 278.1 K throughout the replacement procedure in cases 1–3. **a** The changes in pressure and temperature in case 2; **(b–d)** the variations in the CH₄ compositions in the gas phases at the top and bottom in cases 1–3 (**b**: Case 2-R1-II, **c**: Case 1-R1-II, **d**: Case 3-R1-II); **(e)** the evolution of the top gas composition of CH₄ in the residual gas phase at 278.1 K during the replacement procedure in cases 1–3; **(f)** their P_g·CO₂, P_g·CO₂ of Peq of CH₄-CO₂ mixed gas hydrate, CH₄ hydrate decomposition, and R.Fr.CH₄ in cases 1–4

Table 5 Summary of the efficiency of replacement, $R.F.$, $S.F.$, $R.F.$.

EXP. No	End residual gas of replacement experiment				Efficiency of replacement		
	Partial pressure of CH ₄ (MPa)	Partial pressure of CO ₂ (MPa)	Equilibrium pressure of CH ₄ hydrate (MPa)	Partial pressure CO ₂ of equilibrium pressure of CH ₄ -CO ₂ mixed gas hydrate (MPa) ^a	$R.F.$, CH ₄ (mol%) ^b	$S.F.$, CO ₂ (mol%)	$R.F.$
Case 1-R1-II	2.78	1.05	4.24	1.29	69.30	50.06	2.26
Case 2-R1-II	3.07	0.82	4.24	1.22	84.12	53.23	6.03
Case 3-R1-II	3.30	0.70	4.24	1.07	87.92	44.17	5.76
Case 4-R1-II	2.62	2.04	4.24	1.56	58.47	54.53	1.69
Case 5-R1-III	2.47	1.96	4.24	1.52	58.44	52.09	1.53
Case 6-R1-I	2.57	1.82	4.24	1.60	62.11	55.23	2.02

^a The P^{eq} of CH₄-CO₂ mixed gas hydrate was calculated from the gas composition from the mixed CO₂-CH₄ hydrate decomposition using CSMHyd. The partial pressure CO₂ of the P^{eq} of the CH₄-CO₂ mixed gas hydrate was obtained by multiplying $y_{H_2O}^{CO_2}$ by the P^{eq} of CH₄-CO₂ mixed gas hydrate

^b $R.F.$, CH₄ was calculated using the moles of CH₄ in the residual gas flowing out of the reactor through the mass flow meter. The discrepancy between the consumed gas moles during CH₄ hydrate formation and the total moles of CH₄ flowing out of the reaction through the mass flow meter equaled the difference in mole amounts between the CO₂ moles in the hydrate and the CH₄ moles in the residual gas for cases 1–6. This is because both sets of moles were equal to the CH₄ moles in the vented gas for GC, and 1 mol of CH₄ was replaced by 1 mol of CO₂

3.2.3 Analysis of the effect of $P_g \cdot CO_2$ on replacement reaction

The $P_g \cdot CO_2$ was significantly higher than $P_g \cdot CO_2$ of the P^{eq} of the CH₄-CO₂ mixed gas hydrate when there was no pure CH₄ hydrate decomposition in case 4 (Tables 2 and 4, Fig. 5f). However, in cases 1–3, the $P_g \cdot CO_2$ was lower than $P_g \cdot CO_2$ of the P^{eq} of the CH₄-CO₂ mixed gas hydrate due to the decomposition of pure CH₄ hydrate (Fig. 5f). This indicates that the decomposition of pure CH₄ hydrate facilitated CO₂ sequestration in clathrate and reduced the required $P_g \cdot CO_2$. In molecular dynamics simulation studies, it was observed that methane hydrate decomposition can form semi-clathrate water clusters [44]. Compared to free water, such semi-clathrate water

clusters are more prone to form complete CO₂ hydrate Broussonetia papyrifera structures, thereby reducing the required $P_g \cdot CO_2$.

It should be noted that case 2 was repeated, and after the replacement was completed, the CH₄-CO₂ mixed gas hydrate was heated to 298.1 K without releasing the gas phase, and then cooled to 278.1 K for reforming the CH₄-CO₂ mixed gas hydrate. Surprisingly, no CH₄-CO₂ mixed gas hydrate reformed again. It means that $P_g \cdot CO_2$ was lower than $P_g \cdot CO_2$ of the P^{eq} of the CH₄-CO₂ mixed gas hydrate, thereby terminating the replacement reaction in cases 1–3.

In a summary, combustion heating leads to the decomposition of pure CH₄ hydrates, effectively increasing the

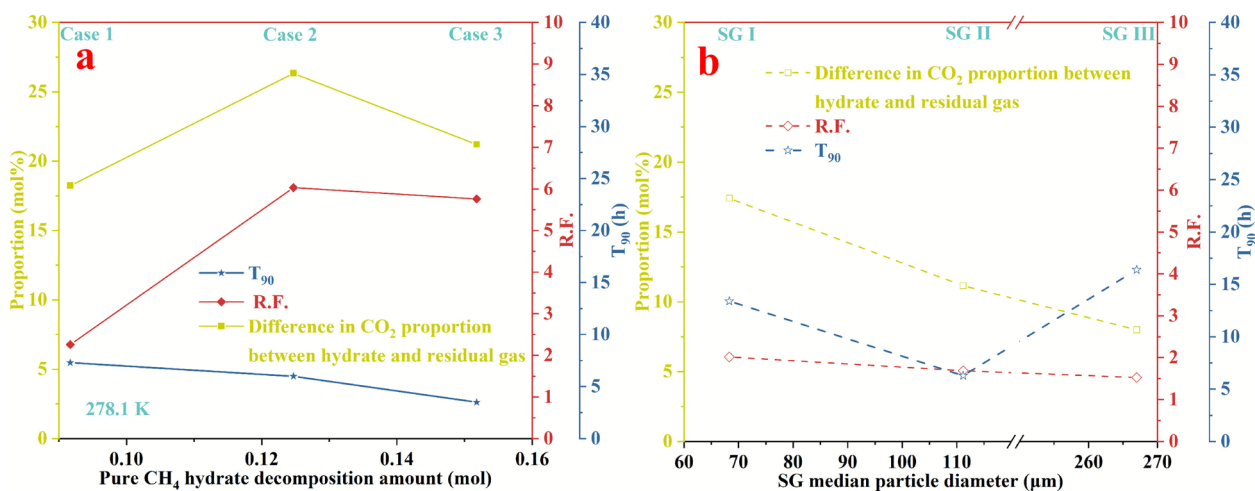


Fig. 6 The effects of CH₄ hydrate particle diameter and pure CH₄ hydrate decomposition amount on replacement efficiency (R.F.), T_{90} , and the difference in CO₂ composition between the hydrate and the gas phase. **a** pure CH₄ hydrate decomposition amount; **b** CH₄ hydrate particle diameter

CH₄ proportion and pressure of extracted gas, thereby enhancing industrial gas extraction efficiency.

3.3 The influence of the SG particle diameter on the replacement efficiency

The SG I, SG II, and SG III particles were employed to synthesize CH₄ hydrate and subsequently perform CO₂-CH₄ replacement experiments at 278.1 K in R1. Following replacement reaction initiation, temperature and pressure stabilized within 3–4 h (Fig. 7d and S3a-c), while the CH₄ proportion in the gas phase exhibited rapid elevation within 24 h followed by gradual stabilization (Fig. 7a-c). Notably, larger hydrate particles demonstrated extended CH₄ hydrate formation duration (Fig. 7f), indicating the existence of a diffusion barrier impeding CH₄ migration from the gas phase to the SG core. This phenomenon implies the absence of interstitial spaces between SG particles and aqueous phase for CH₄ penetration. The diameter of CH₄ hydrate particles has an extremely small effect on the gas phase CH₄ proportion and the rate of pressure increase during the replacement process, as well as on parameters such as the $R.Fr.CH_4/S.Fr.CO_2$, terminal pressure values, CO₂ proportions in hydrate (Fig. 7d-f), the difference of CO₂ proportion between in the hydrate and the gas phase and the R.F. (Fig. 6b).

3.4 CH₄ hydrate decomposition during replacement procedure

3.4.1 CO₂ liquefaction, gasification, CH₄ hydrate decomposition, replacement, and mixed gas hydrate decomposition stage division

Changes in pressure, temperature, and heat flow during the CH₄-CO₂ hydrate replacement with 0.148 mol/mol CO₂ injection (case 7-R2-II in table S1) are shown in Fig. 8. Figure 8a shows CH₄ release and CO₂ injection starting at Stage A (T < 233.1 K) with subsequent CO₂ liquefaction. Due to liquid nitrogen cooling, the DSC cell temperature couldn't rise immediately post-A. The phase transition from liquid to gaseous CO₂ unfolded gradually from point B to point C during Period 1. (223.1–245.3 K). Period 2's pressure increase mainly resulted from thermal gas expansion. The sharp pressure surge in Period 3 indicated CH₄ hydrate dissociation, aligned with Stage D's endothermic peak. Finally, the endothermic peak at point E reflected CH₄-CO₂ mixed gas hydrate decomposition during heating. Pressure and heat flow stabilized during 278.1 K constant temperature phase, indicating complete replacement without oscillating heat flow (Fig. 8a). Figure 8b presents the pressure-temperature profiles at critical stages (A, B, C). Points A/B's pressure ≥ CO₂ gas-liquid equilibrium pressure (P^{eq}), which prevented the CH₄ hydrate from decomposing violently. Point C's

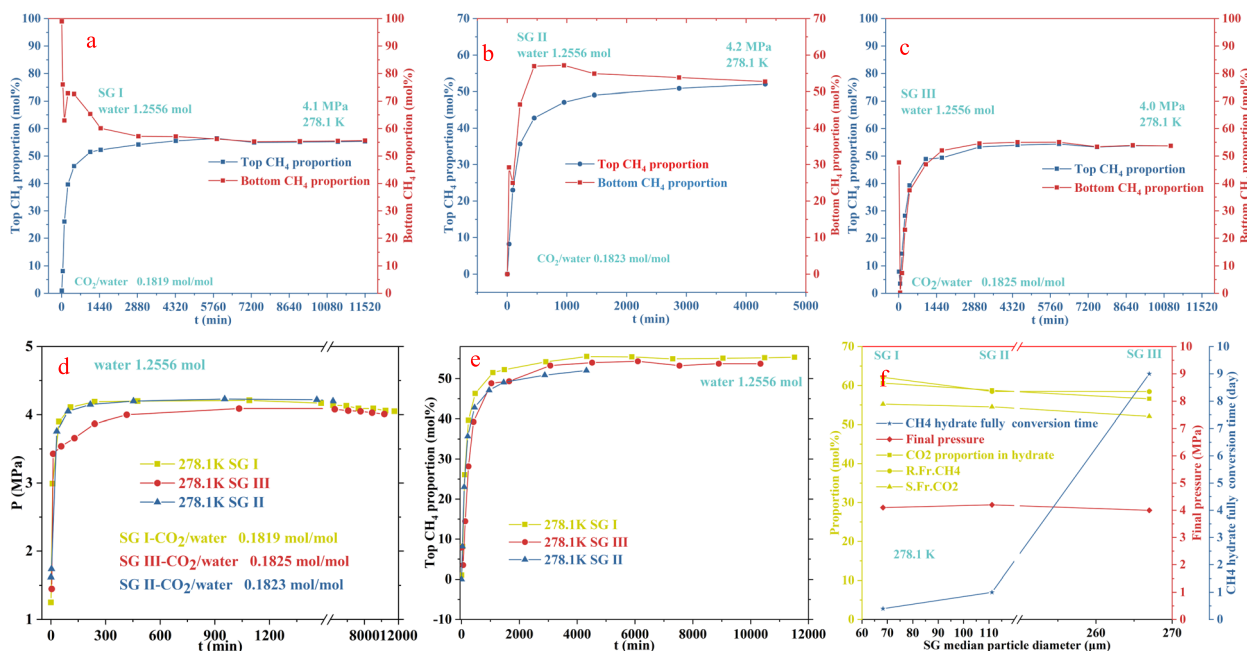


Fig. 7 a-c Variations in CH₄ proportion within gas phases at top and bottom positions across cases 9–11 (a: Case 11-R2-I, b: Case 9-R2-II, c: Case 10-R2-III); (d-e) pressure fluctuations and top-layer CH₄ composition changes observed in cases 9–11 (d: pressure profile, e: CH₄ proportion at top); (f) comparative parameters including $R.Fr.CH_4/S.Fr.CO_2$, CH₄ hydrate formation duration, terminal pressure values, and CO₂ proportion in hydrate across cases 9–11

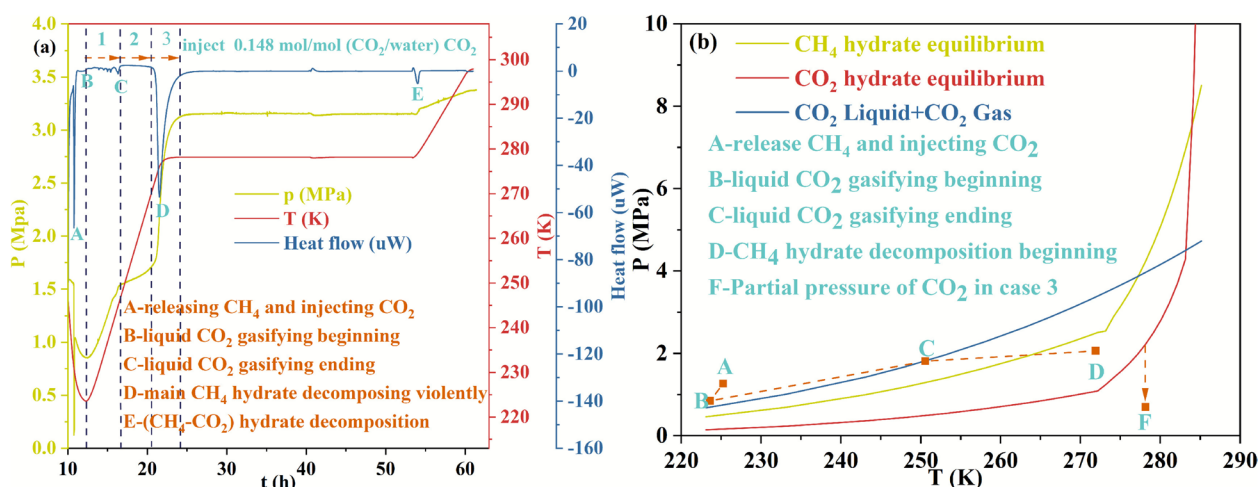


Fig. 8 The pressure, temperature, and heat flow variations during the CH₄-CO₂ hydrate replacement experiment (case 7-R2-II in table S1).

(a-1) depicts the procedure of liquid CO₂ vaporization, (a-2) illustrates the CH₄-CO₂ hydrate replacement process, (a-3) shows the CH₄ hydrate decomposition process, (a-E) mixed gas hydrate decomposition; (b) displays the pressure and temperature at time points A, B, C, and D in the DSC replacement experiment (case 7-R2-II in table S1); (F) represents the partial pressure of CO₂ in case 3

pressure was between CH₄ and CO₂ hydrate phase equilibrium pressures.

Therefore, when utilizing DSC to analyze replacement experiments, it is essential to first identify the replacement stage before analyzing whether CH₄ hydrate decomposes during the replacement process using the heat flow curve.

3.4.2 Analysis of the heat flow and pressure during the replacement and pure CH₄ hydrate decomposition stage

Figure 9 shows pressure, temperature, and heat flow variations during CH₄-CO₂ hydrate replacement with CO₂/water ratios of 0.278–0.313 mol/mol. At 228 K/1.0 MPa (point A), CO₂ liquefaction occurred (case 8-R2-II in table S1). During period 2 (Fig. 9a), a broad endothermic peak from CH₄ hydrate decomposition appeared, while multiple small exothermic reactions superimposed due to CO₂ hydrate formation between 258.1–268.1 K. This confirms that CO₂ hydrate formed during pure CH₄ hydrate decomposition. At this point, the decomposition rate of CH₄ hydrate exceeded the formation rate of CO₂ hydrate. In period 3, decreased CH₄ decomposition rate (caused by rising $P_g \cdot \text{CH}_4$) dissipated the broad peak. Both CH₄ decomposition (small endothermic peak) and CO₂ formation (small exothermic peak) occurred simultaneously at matching rates (1:1 replacement without pure CH₄ decomposition), creating horizontal heat flow oscillations. This aligns with reported similar formation heats for CH₄ and CO₂ hydrates [25]. This indicates hydrate particles decomposed during replacement,

consistent with simulations and Raman analyses [43–45] but contradicting conventional DSC results [49].

Figures 9b and 9c show the overall pressure increased at 278.1 K during replacement, aligning with CH₄/CO₂ Z values' influence on gas phase pressure [52, 53, 65, 66]. Figure 9c shows the pressure trend rose steadily at 278.1 K during period 6, indicating 1 mol CH₄ was replaced by 1 mol CO₂ in hydrates, consistent with CH₄-CO₂ replacement mechanism in hydrate large cages [65, 66]. The horizontal heat flow line aligned with similar formation specific heats for CH₄ and CO₂ hydrates [25]. This indicates that the time period 6 shown in Fig. 9c was the stage of replacement reaction, during which CH₄ hydrate decomposition occurred.

In cases 2–5, closely spaced oscillation periods occurred during the transition from CH₄ hydrate decomposition to replacement in heat flow curves (Fig. 9d), indicating the endothermic peak was due to CH₄ decomposition. The oscillation period ranged from 0.11 to 0.21 h, and the oscillation duration ranged from 0.6 to 13.8 h.

In a summary, CH₄ hydrate decomposed during replacement reaction procedure, and CO₂ hydrate formed during pure CH₄ hydrate decomposition procedure.

3.5 Variation in replacement efficiency across depths with varying spherical hydrate particle diameters

In different positions and various cases, the peak area ratio of large-cage to small-cage CH₄ hydrate was almost identical in pure CH₄ hydrate (Fig. 10a-b). The peak area ratio of large-cage to small-cage CH₄ hydrate at position 5 measured 2.16, lower than the 3.49 ratio observed in pure CH₄ hydrate (Fig. 10c), indicating

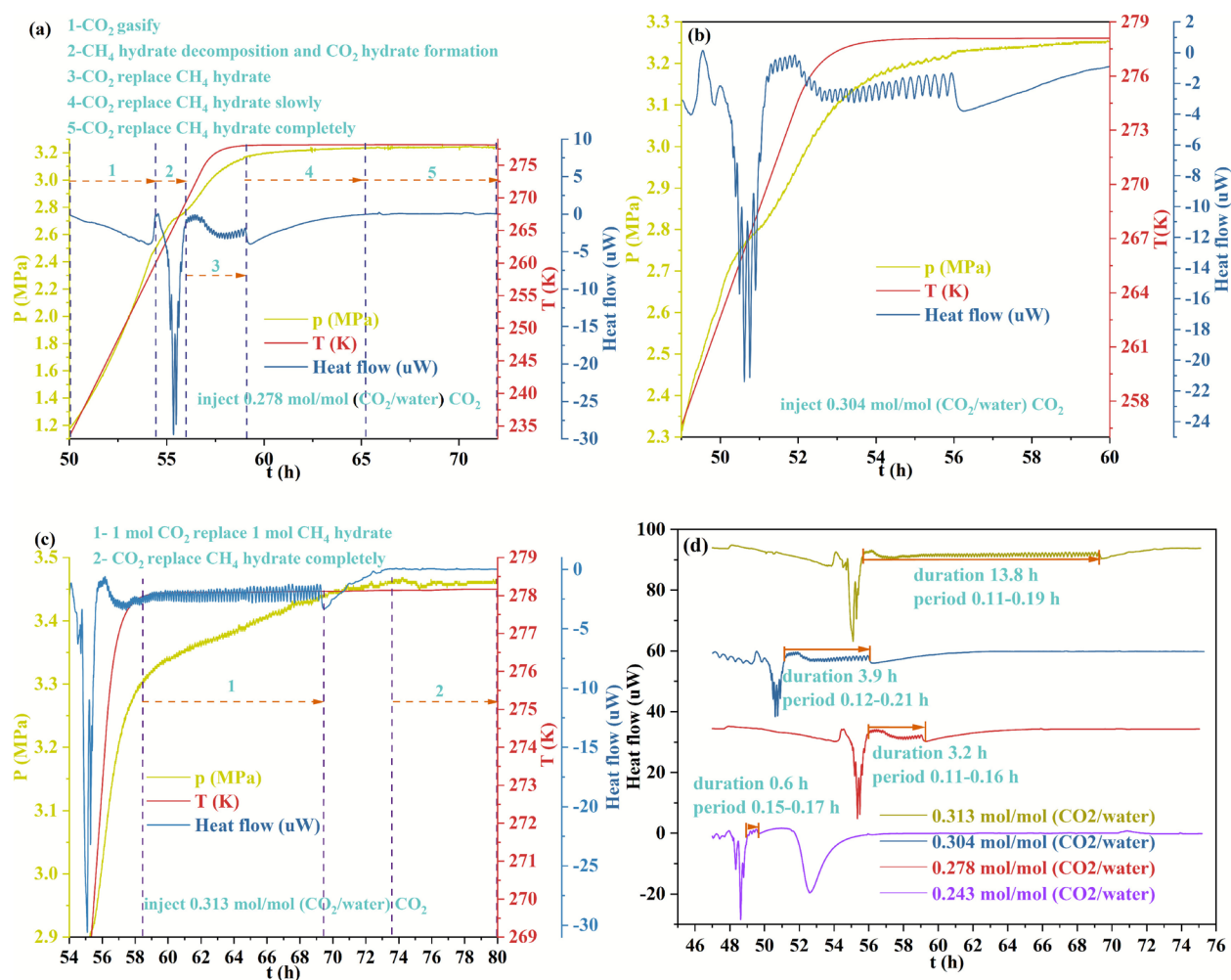


Fig. 9 Pressure, temperature and heat flow change for the CH_4 - CO_2 hydrate replacement experiment and CH_4 - CO_2 mixed gas hydrate formation experiment using the decomposition gas of the CH_4 - CO_2 hydrate replacement experiment with 0.278 mol/mol (CO_2 /water) CO_2 injection. **a** Inject 0.278 mol/mol (CO_2 /water) CO_2 ; **b** inject 0.304 mol/mol (CO_2 /water) CO_2 ; **c** inject 0.313 mol/mol (CO_2 /water) CO_2 ; **d** changes in heat flow curves for CH_4 - CO_2 hydrate replacement experiment with different masses of injection CO_2

greater CO_2 replacement efficiency in large cages [67]. Position 6 showed a reduced ratio of 1.12 compared to position 5, demonstrating more extensive large-cage replacement (Fig. 10c-d). While position 7 exhibited a CH_4/CO_2 hydrate intensity ratio of 2.70 versus position 8's 2.55, this minor variation suggests comparable replacement efficiency between these locations (Fig. 10e).

Figure 11f shows the peak area ratio variation of large/small cage CH_4 hydrate from surface to center in case 4 after replacement. The ratio fluctuated between 1.11–1.15 (case 4 position 6) and 1.64–1.66 (case 4 position 9) from surface to center with minimal variation. This indicates similar CO_2 replacement ratios for large cage CH_4 in hydrate particle centers compared to other positions in SG II.

(See figure on next page.)

Fig. 10 **a-b** The Raman spectra of pure CH_4 hydrate in different cases and at different locations (**a**: different locations, **b**: different cases); **c** Raman spectra correspond to CH_4 - CO_2 hydrate at positions 5 and 6 after replacement at a temperature of 278.1 K in case 4 (water 1.2556 mol, CO_2 /water 0.1823 mol/mol) and pure CH_4 hydrate; **d** a picture of spherical hydrate particle on SG in a Raman microscope; **e** Raman spectra correspond to CH_4 - CO_2 hydrate at positions 7 and 8 after replacement at a temperature of 278.1 K in case 4 (water 1.2566 mol, CO_2 /water 0.1823 mol/mol); **f** the CH_4 ratio in the large cage to the small cage changes from the spherical hydrate particle surface to center in case 4

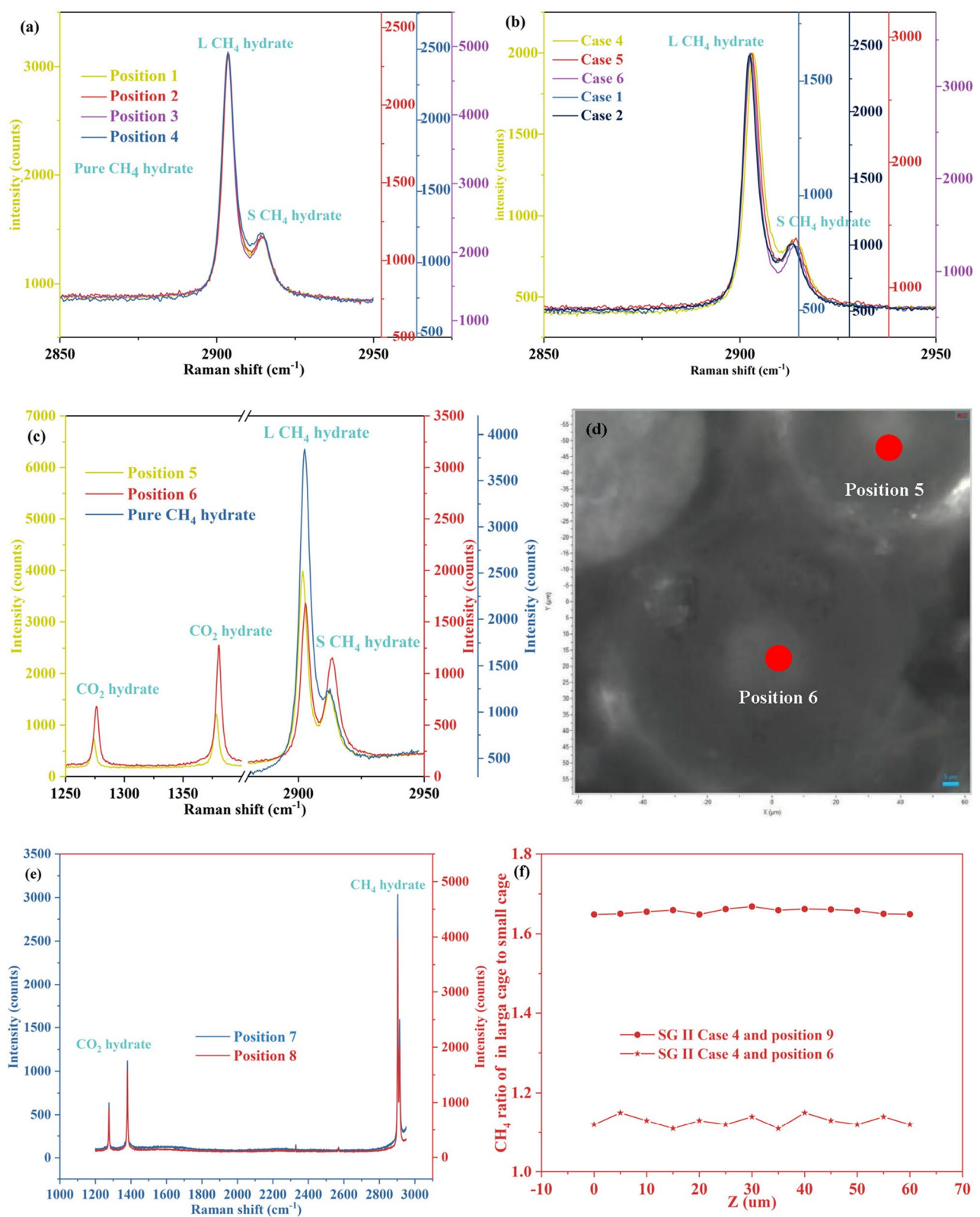


Fig. 10 (See legend on previous page.)

In summary, due to the decomposition of CH_4 hydrate during the replacement process, it may have provided a pathway for the diffusion of CO_2 , resulting in the same replacement efficiency at different depths of hydrate particles. Therefore, under the same conditions, the diameter of the hydrate particles has no effect on the replacement rate and efficiency. The entire replacement process belongs to the reaction-controlled stage.

4 Conclusions

When SG pore was fully saturated with water and completely converted into ice and CH_4 hydrate, volumetric expansion caused a compact ice or hydrate layer to form on its surface, producing stable spherical particles. Throughout hydrate dissociation, water gradually returned to the material's pores.

The CH_4 hydrate decomposed during the replacement process and CO_2 gas hydrate could form during the CH_4 hydrate decomposition process. The decomposition of pure CH_4 hydrate facilitated the easier engagement of CO_2 into the hydrogen-bonded water cage, resulting in the $P_g \cdot \text{CO}_2$ falling below the $P_g \cdot \text{CO}_2$ of the P^{eq} of the CH_4 - CO_2 mixed gas hydrate and then halting the replacement reaction.

The diameter of the spherical hydrate particles had little practical influence on the replacement rate and amount. The ratio of replaced large cage CH_4 hydrate with CO_2 in the center of the spherical hydrate particles was almost identical to that at other positions in the SG. The replacement reaction was controlled by reaction kinetic in the SG.

In the present study, the proportion of CH_4 in the residual gas was enhanced to 84.67 mol%. Simultaneously, the $S.Fr.\text{CO}_2$ reached 44.17 mol%.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s43979-026-00161-2>.

Supplementary Material 1.

Authors' contributions

Jun Liu: Conceptualization, Writing—original draft, Investigation, Formal analysis, Validation, Visualization. Rongsheng Lin: Formal analysis, review & editing, Validation. Zhaoqi Zheng: Formal analysis. Hailong Zhang: Formal analysis. Bingyuan Hong: review & editing. Jianfeng Gao: review & editing. Bingyuan Hong: review & editing. Deqing Liang: formal analysis, Validation. Zhenyuan Yin: formal analysis. Jinlong Zhu: Formal analysis, review & editing, Validation. Yusheng Zhao: Supervision, Validation, Project administration.

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Data availability

Data and materials will be made available on reasonable request.

Declarations

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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